

Super Evolution Theory:

From Receptor-State Computation to Reticulate Evolution

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August 9, 2026

Abstract

A chemical signal does not carry a complete biological instruction by itself, and its work does not begin or end at receptor binding. Serotonin must be synthesized, loaded into vesicles, released, distributed through extracellular space, detected by a receiver, transformed through receptor-specific pathways, limited by feedback, retrieved or metabolized, and followed by the recycling or replacement of both signaling and receiving machinery. The same concentration can therefore produce different consequences when receptor subtype, abundance, localization, conformation, RNA editing, membrane voltage, intracellular partners, metabolic condition, exposure history, or clearance differs.

Super Evolution Theory (SET) joins these distinguishable operations into a receiver-centered account of cellular and evolutionary change. At the cellular scale, a receptor ecosystem selects and transforms signals; electrical, biochemical, metabolic, and rhythmic state determines which routes are available; feedback and trafficking alter future sensitivity; and repeated activity can change expression or structure. SET calls its proposed reconstruction of actionable cellular state from distributed, time-structured differences **Cellular Oscillating Tomography (COT)**. At the evolutionary scale, mobile elements, plasmids, viruses, transformation, conjugation, transduction, and symbiotic combination can move already assembled biological modules across boundaries. SET calls this **packetized evolution** only when a bounded carrier, transferable content, transport route, receiver compatibility, and measurable consequence are specified.

The serotonin system supplies a worked mechanism rather than a decorative example. Betty Twarog and Irvine Page established serotonin in mammalian tissues including brain; Diego Walther and colleagues later separated central synthesis through the TPH2 isoform. Bernard Katz's release-timing problem, John Heuser and Thomas Reese's vesicle-recycling evidence, Mary Hare Bernheim's monoamine-oxidase discovery, Earl Sutherland's second-messenger logic, and John Langley's receptor distinction each identify a different causal stage. Most 5-HT receptor classes are G-protein-coupled receptors, whereas 5-HT₃ is a pentameric ligand-gated cation channel. Receptor editing, location, ligand residence, internalization, autoreceptor feedback, SERT-mediated reuptake, MAO-mediated metabolism, and neighboring-cell uptake shape both the signal trajectory and the state left behind. The manuscript also corrects a recurrent terminology error: **5-HTP is a serotonin precursor; 5-HT receptors are the receptor family**.

SET does not replace mutation, drift, recombination, development, natural selection, receptor pharmacology, robust adaptation, or established horizontal gene transfer. It proposes that these mechanisms should be measured in one stateful and reticulate architecture. The manuscript preserves the receptor evidence atlas, complete signal lifecycle, pharmacological perturbation program, cellular-memory architecture, carrier-by-timescale model, evolutionary-network account, and executable synthetic model. It also supplies a pinned Lean 4 formal kernel. Fifteen machine-checked statements prove finite-path preservation and reachability monotonicity under edge extension, nonnegativity of the

six-factor transfer rate, nonincrease of reciprocal expected waiting time under a nonnegative independent added hazard, response-domain and carrier identities, and an exact rational counterexample in which a newly supported detour accompanies an increase in expected hitting time from 2 to 2,600. The counterexample is as important as the favorable theorem because it prevents “more routes” from being mistaken for “faster evolution” without transition probabilities and costs. These are proofs of consequences inside stated mathematical models, not empirical confirmation that living systems instantiate SET.

1 Begin with an ordinary biological problem

Imagine exposing a cell to the same concentration of the same signaling molecule twice.

During the first exposure, the relevant receptor may be present at the membrane, a downstream enzyme may be available, the cell may be near an electrical threshold, and sufficient metabolic resources may exist to complete the response. During the second exposure, some receptors may have been internalized, an autoreceptor may have reduced additional transmitter release, calcium stores may differ, a transporter may shorten the signal, a gene-expression program may have changed, or the cell may occupy another phase of an electrical, metabolic, or biochemical cycle.

The input molecule is nominally unchanged. The operation is not.

This produces the first problem addressed by SET:

Where does the functional meaning of a biological signal reside when the same signal produces different consequences in different receivers, or in the same receiver at different times?

The answer cannot reside in the ligand alone. Functional meaning is produced by the relation among the signal, the receiver’s current organization, the receiver’s history, and the routes that remain available when the signal arrives.

Now consider an evolutionary problem. A lineage can acquire a functional module assembled elsewhere. A bacterium receiving a plasmid does not need to reconstruct every intermediate through which the donor’s module arose. The recipient must still encounter the carrier, admit or retain it, express it, survive its costs, and pass selection. Nevertheless, the transition graph has changed: a route now exists that a lineage-only account omitted.

This produces the second problem:

How should evolutionary search be represented when bounded biological modules can move between organisms or lineages and be reused?

SET connects these problems. The cell is a stateful receiver. Evolution acts on populations of such receivers while also redistributing parts of their control architecture.

2 The receiver idea has a scientific history

John Newport Langley’s early-twentieth-century idea of cellular “receptive substances” matters because it separates the chemical from the apparatus that makes the chemical consequential [1]. In plain language, a

drug or transmitter does not reach into a cell and issue an abstract command. It acts through differentiated cellular material able to receive it.

Langley's name earns its place here by defining the paper's starting distinction. Alan Hodgkin and Andrew Huxley later made electrical receiver state quantitative: membrane response depends on voltage and state-dependent conductances, not on an abstract all-or-none label alone [5]. François Jacob and Jacques Monod showed how environmental conditions can be connected to altered protein synthesis through physical regulatory machinery [6]. Jacques Monod, Jeffries Wyman, and Jean-Pierre Changeux formalized how ligand binding, protein conformation, and cooperativity can belong to one allosteric system [7]. Conrad Waddington's developmental landscape supplied a durable picture of path dependence: the routes available next depend partly on how the system arrived at its current state [8].

These landmarks do not prove SET. They establish the scientific pieces that any credible receiver-centered theory must inherit. SET's burden is to show whether joining the pieces yields additional predictive power.

3 The established account comes before the SET terminology

Cells synthesize receptors, fold them, inspect them through quality-control systems, traffic them through intracellular compartments, insert them into membranes, cluster them, modify them, internalize them, recycle them, or degrade them. Ligands bind with finite affinity and imperfect selectivity. Ionotropic receptors can directly change conductance. Metabotropic receptors can recruit G proteins, enzymes, second messengers, ion channels, transcriptional regulators, and structural processes. Transporters move ligands across membranes. Enzymes synthesize and metabolize them. Feedback changes release, gain, and availability.

The established scientific picture already implies a causal sequence:

```
signal identity, concentration, and timing
-> receptor class, subtype, and location
-> receiver configuration and prior state
-> physical transduction route
-> intracellular and cellular consequence
-> feedback, termination, trafficking, and persistence
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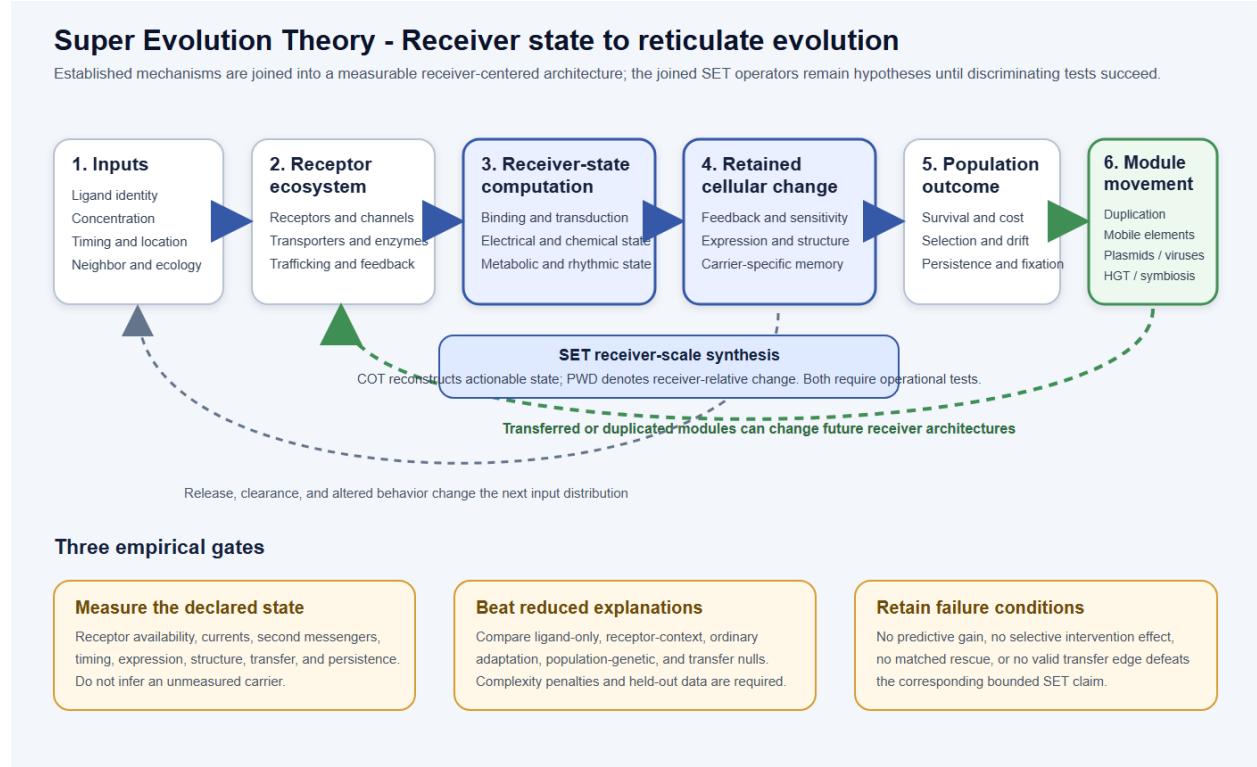


Figure 1: Receiver-state architecture linking environmental input, receptor ecosystems, cellular computation, retained change, population outcome, and reticulate module movement. The lower paths identify feedback and the three empirical gates required before a joined SET mechanism is supported.

SET enters only after that sequence is understood. It proposes that the sequence should be treated as a configurable cellular input-and-control surface rather than as isolated facts assigned to separate subfields.

4 The receiver must be part of the signal model

Let the state of cell (i) at time (t) be

$$x_i(t) = (r_i(t), m_i(t), q_i(t), o_i(t), g_i(t), e_i(t)), \quad (1)$$

where:

- (r_i) is receptor, channel, transporter, scaffold, and trafficking configuration;
- (m_i) is membrane voltage, ionic concentration, and conductance state;
- (q_i) is intracellular biochemical and second-messenger state;
- (o_i) is oscillatory, rhythmic, or temporally organized state;
- (g_i) is transcriptional, epigenetic, regulatory, and genetic state;
- (e_i) is energetic and metabolic state.

An arriving signal (s_i(t)) becomes an effective intracellular drive:

$$u_i(t) = F_i(s_i(t), x_i(t), \eta_i(t)), \quad (2)$$

where $(\eta_i(t))$ contains fluctuations and unresolved variables. The next cell state depends on the prior state, transformed input, neighboring cells, and environment:

$$x_i(t + \Delta t) = G_i(x_i(t), u_i(t), \mathcal{N}_i(t), E(t)). \quad (3)$$

Equations (1)-(3) do not assert that every cell uses the same variables. They prevent a specific error: treating a ligand as a context-free instruction.

The word **computation** is used operationally. A receiver discriminates among inputs and internal conditions, transforms them through physical mechanisms, and changes its future state. No digital symbols, central processor, or conscious awareness are required.

5 From receptors to a receptor ecosystem

A receptor list is not yet a system. To predict an outcome, one may need the receptor's class, subtype, abundance, cellular and subcellular coordinates, conformation, post-transcriptional form, subunit composition, phosphorylation state, coupling partners, and recent exposure. One may also need transporters, degrading enzymes, membrane voltage, ion gradients, organelles, neighboring cells, and feedback routes.

Represent the effective availability of receptor channel (k) in cell (i) by $(a_{ik}(t))$, the ligand environment by $(\mathbf{L}(t))$, and binding plus proximal activation by (B_k) :

$$z_{ik}(t) = a_{ik}(t)B_k(\mathbf{L}(t), \theta_{ik}(t)), \quad (4)$$

where $(\theta_{ik}(t))$ contains affinity, conformation, location, coupling, inhibition, and regulatory context. The downstream response is a nonlinear composition:

$$\mathbf{u}_i(t) = \mathcal{C}_i(\mathbf{z}_i(t), x_i(t), \mathcal{N}_i(t)). \quad (5)$$

Only after this operation is clear is the SET term useful. A **receptor ecosystem** is the interacting set of receptors, channels, transporters, enzymes, scaffolds, ligands, cellular compartments, feedback loops, and neighboring cells that jointly determine how a signal is transformed. The word “ecosystem” identifies interdependence and stateful interaction; it does not claim that every receptor system should be described by ecological population equations.

6 One transmitter can address qualitatively different operations

Serotonin is a useful worked example because it defeats the idea that a transmitter name specifies one message.

NC-IUPHAR classifies the major vertebrate 5-HT receptors as GPCRs with one decisive exception: 5-HT₃ is a cation-selective pentameric ligand-gated ion channel [11,12]. David Julius and colleagues cloned and functionally expressed 5-HT₃ as a serotonin-gated channel [13]. Derkach, Surprenant, and North

independently established membrane-ion-channel behavior [32]. A 5-HT₃-bearing receiver can therefore translate serotonin into rapid conductance and depolarization. A GPCR-bearing receiver instead initiates metabotropic cascades whose effects unfold through cellular machinery.

Within the GPCR families, the principal routes also differ:

Table 1: Principal operational routes of major 5-HT receptor families. These are summary couplings, not exclusive outputs.

Family	Principal route	Plain-language operational contrast
5-HT ₁ and 5-HT _{5A}	Gi/o	Often reduces cAMP-linked activity or transmitter release, depending on location and partners
5-HT ₂	Gq/11	Commonly recruits PLC-linked signaling and downstream calcium or channel effects
5-HT ₄ , 5-HT ₆ , and 5-HT ₇	Gs	Commonly increases adenylyl-cyclase/cAMP-linked signaling
5-HT ₃	Direct ion conductance	Opens a cation channel rather than beginning with a G-protein cascade

These are principal-coupling summaries, not claims of exclusive output. Additional partners, receptor reserve, expression system, ligand, exposure duration, and cell type can change the observed trajectory.

The scientific consequence is already stronger than “serotonin has many effects.” The same chemical environment can address different physical operations because different cells expose different receiver architectures.

7 Receptor diversity forms a conditional input alphabet

Let a cell’s serotonin receptor-state vector be

$$\mathbf{r}_i(t) = (a_{1A}, a_{1B}, a_{1D}, a_{1E}, a_{1F}, a_{2A}, a_{2B}, a_{2C}, a_3, a_4, a_{5A}, a_6, a_7)_i. \quad (6)$$

This vector is not a count of gene names. Each entry represents effective availability and route accessibility. The human 5-HT_{5B} locus supplies a caution: it is classified as a pseudogene, so a cross-species receptor table must not silently treat it as a functional human channel [11].

Receptor diversity can enlarge the set of discriminable input-response relations, but additional subtypes do not guarantee additional information. Cross-reactivity, saturation, correlated expression, shared downstream pathways, noise, energetic cost, and limited readout can make apparently different channels redundant.

For input condition (S) and measured response (U), the appropriate quantity is not the number of receptor labels but the mutual information

$$I(S; U) = \sum_{s,u} p(s, u) \log_2 \frac{p(s, u)}{p(s)p(u)}. \quad (7)$$

Claude Shannon’s work belongs here because this is a quantitative discrimination problem [2]. Shannon does not supply biological meaning on his own. The receiver, its action, and the reproducible consequence

provide causal semantics.

SET prediction: in a regime where receptor diversity is active, a measured receptor-state vector should improve held-out prediction or input-response mutual information relative to ligand concentration and cell label alone.

Failure condition: after penalizing model complexity, receptor-state variables add no reproducible predictive information.

8 RNA editing demonstrates receiver-dependent routing

The 5-HT_{2C} receptor is the strongest current example of the receiver being changed before the next ligand arrives.

Colleen Burns and colleagues found tissue-specific edited receptor isoforms and reported that extensively edited forms could reduce receptor-G-protein coupling efficiency by approximately an order of magnitude in their experimental system [14]. Kelly Berg and colleagues compared edited and non-edited isoforms at matched expression and found approximately tenfold differences in serotonin potency together with altered agonist routing between phospholipase-C/inositol-phosphate and phospholipase-A₂/arachidonic-acid responses [15].

The exact numerical effects belong to those preparations; they are not universal constants. The bounded result is nevertheless decisive for the SET architecture:

```
same nominal endogenous ligand
-> different edited receptor configuration
-> different coupling efficiency and effector routing
-> different cellular response
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Changing the receiver changes the operation performed on the signal.

RNA editing is not automatically a DNA mutation, not necessarily permanent, and not by itself transgenerational inheritance. Its relevance here is narrower and stronger: cellular input semantics depend partly on configurable receiver structure.

9 Receptor location supplies coordinates to the alphabet

A receptor identity without a location is an incomplete causal description.

The 5-HT_{1A} subtype can participate in somatodendritic autoreceptor feedback in serotonergic raphe neurons and in heteroreceptor responses in target populations. Terminal 5-HT_{1B} autoreceptors can inhibit additional serotonin release in specified projection regions [18,19]. The same named receptor family therefore occupies different loops depending on its cellular coordinate.

Vincent Beliveau and colleagues produced a high-resolution in vivo atlas of selected components of the human serotonin system, demonstrating large regional differences in receptor and transporter density [17]. Bounded single-cell studies add a second scale: individual neurons in a specified population can express 5-HT_{1A}, 5-HT_{2A}, 5-HT_{2C}, and 5-HT₇ alone or in combinations associated with distinct response profiles [33].

These sources do not support an exhaustive whole-body map. They do establish that spatial distribution and co-expression are not optional detail.

SET prediction: a model containing presynaptic, somatodendritic, postsynaptic, surface, intracellular, region, and cell-class coordinates should outperform a receptor-count model where localization is causal.

Failure condition: localization adds no reproducible predictive value after receptor abundance and cell identity are controlled.

10 Ligand kinetics and receptor conformation add temporal semantics

Binding is a trajectory, not a timeless switch.

David Wacker and colleagues' LSD-bound serotonin-receptor structures and functional experiments connected extracellular-loop structure and ligand residence to time-dependent signaling at 5-HT₂ receptors [16]. NC-IUPHAR also records receptor-dependent differences in arrestin preference and notes that 5-HT receptors can interact with partners beyond their principal G proteins [11].

This creates two requirements for SET:

1. affinity alone cannot stand in for the complete downstream trajectory;
2. assigning a receptor to Gi/o, Gq/11, or Gs cannot be treated as an exclusive path under every ligand, duration, and cellular state.

The relevant input includes identity, concentration, sequence, duration, and residence. The receiver includes conformation, accessible partners, location, and history. Their interaction produces a time course.

This is also why drugs are scientifically valuable perturbations. Agonists, partial agonists, inverse agonists, antagonists, reuptake inhibitors, and other compounds can expose otherwise hidden branches in the control architecture. A pharmacological perturbation is evidence about a mechanism only when target engagement, concentration, preparation, timing, and off-target limits are specified.

11 The receiver is altered by receiving

Receptor availability changes through synthesis, insertion, clustering, modification, desensitization, internalization, recycling, and degradation. Schmid and colleagues demonstrated dynamin-dependent 5-HT_{2A} internalization and distinguishable trafficking under specified agonist and antagonist conditions [20]. The same external serotonin concentration can therefore encounter a different surface after prior exposure.

A minimal availability model is

$$\frac{da_{ik}}{dt} = J_{ik}^{\text{insert}} + J_{ik}^{\text{recycle}} - J_{ik}^{\text{internalize}} - J_{ik}^{\text{degrade}}, \quad (8)$$

where each flux can depend on ligand history, phosphorylation, arrestin or other partners, membrane traffic, synthesis, and cell state.

This changes the central SET argument. Receptor state is not merely an initial condition. Signal interpretation feeds back into maintenance of the interpreting apparatus.

Prediction: matched cells given different exposure histories should show measurable differences in receptor surface availability and later response.

Failure condition: the apparent history effect disappears when abundance, localization, and other identified variables are controlled.

12 A signal is a complete biological lifecycle

Consider what must happen before a serotonin molecule can influence a neuron and what must happen before that neuron is ready to interpret another event. The transmitter must be made, stored, released, distributed, received, and ended. The vesicle membrane must be recovered or replaced. The receptor may remain available, become desensitized, move inside the cell, recycle to the surface, or be degraded. The resulting cell may return toward its former state or enter a different one. A theory that begins at receptor binding and stops at “disposal” leaves out much of the computation.

12.1 Synthesis begins before the receptor

Betty Twarog and Irvine Page established serotonin in sampled mammalian tissues, including brain, before modern receptor taxonomy existed [34]. Diego Walther and colleagues later identified a second tryptophan-hydroxylase isoform, TPH2, associated with central serotonin synthesis [35]. The biochemical sequence also fixes an important terminology boundary:

```
tryptophan
-> tryptophan hydroxylase
-> 5-hydroxytryptophan (5-HTP)
-> aromatic L-amino-acid decarboxylase
-> 5-hydroxytryptamine (5-HT, serotonin)
```

5-HTP is therefore a precursor, not a receptor-family name. The appropriate phrases are **5-HT receptor**, **serotonin receptor**, or a specified subtype such as **5-HT2A** [36]. Older source passages using “5-HTP receptor” must be read as terminology to recover, not as a reason to discard the receptor-computation argument.

12.2 Storage, release, and vesicle reuse are separate operations

VMAT2 uses a proton-driven alternating-access cycle to load serotonin and other monoamines into vesicles before release [21]. Bernard Katz and Ricardo Miledi made the timing of calcium action in transmitter release experimentally explicit [38]. Martin Geppert, Thomas Südhof, and colleagues then showed in a mutant central-synapse preparation that synaptotagmin I is a major calcium sensor for fast synchronous release, while several asynchronous components remain [37]. The distinction matters to SET: coordinated release and total release are not interchangeable variables.

John Heuser and Thomas Reese demonstrated that vesicle membrane retrieval and recycling are physical stages of reuse rather than an invisible reset [39]. Vesicle recycling is not receptor recycling. The first restores transmitter-packaging machinery; the second changes the future sensitivity and location of the receiver.

Let $q_v(t)$ denote the readily available vesicle pool:

$$\frac{dq_v}{dt} = J_{\text{refill}}(t) + J_{\text{recycle}}(t) - J_{\text{release}}(t) - J_{\text{vesicle loss}}(t). \quad (9a)$$

Release probability can depend on calcium history, vesicle state, presynaptic inhibition, and the timing relation among arriving inputs. A measured transmitter pulse is thus downstream of a prepared sender state.

12.3 Diffusion creates a spatial and temporal field

After release, transmitter does not become a context-free message delivered to exactly one address. Michael Bunin and R. Mark Wightman quantified serotonin release and uptake in an extrasynaptic-transmission analysis, constraining how diffusion and retrieval shape reach and duration [40]. The local concentration field can be represented provisionally as

$$\frac{\partial c(\mathbf{r}, t)}{\partial t} = D_c \nabla^2 c(\mathbf{r}, t) + J_{\text{release}}(\mathbf{r}, t) - J_{\text{SERT}}(\mathbf{r}, t) - J_{\text{neighbor}}(\mathbf{r}, t) - J_{\text{metabolism}}(\mathbf{r}, t), \quad (9b)$$

where each sink is a separate measurable operation. This equation is deliberately more specific than saying that serotonin is “cleared.” A change in diffusion, transporter density, neighboring-cell uptake, or metabolism changes the spatiotemporal pattern presented to receptors.

12.4 Binding is transformed by the receiver

Binding then enters the receptor ecosystem established in Sections 5-11. The same ligand can open the 5-HT₃ cation channel, inhibit or stimulate second-messenger routes through different GPCR families, act through somatodendritic or terminal autoreceptors, encounter edited 5-HT_{2C} variants, or reach membrane and intracellular receptor populations with different access constraints.

Earl Sutherland’s second-messenger work matters here because it made the bridge from extracellular signal to intracellular consequence experimentally and conceptually explicit [42]. The ligand does not carry cyclic AMP, calcium release, kinase activity, transcription, or morphological change as a miniature instruction. Receptor class, coordinate, conformation, coupling partners, and the pre-existing cell state generate the downstream operation.

Autoreceptors also change the sender. Manipulating raphe 5-HT_{1A} autoreceptor levels changes serotonergic firing and behavioral response in the studied mouse system [18]. Knockout and agonist experiments support a terminal inhibitory role for 5-HT_{1B} in specified regions [19]. Feedback therefore changes both present interpretation and future transmitter supply.

12.5 Ending a signal is an active transformation

Julius Axelrod, Leslie Whitby, and Georg Hertting made transmitter recovery experimentally visible by showing that psychoactive drugs could reduce tissue uptake of radiolabeled norepinephrine [45]. Their result concerns catecholamine uptake rather than the serotonin transporter, so it supplies a historical and mechanistic landmark, not a serotonin-specific substitute. In the serotonin system, SERT uses ion gradients and alternating conformations to retrieve serotonin from extracellular space [22]. Structures of SERT bound to antidepressants show how selective serotonin-reuptake inhibitors occupy the central site and stabilize a

transporter state that blocks ordinary transport [23]. Reuptake is not degradation: transporter action changes where the molecule is located, whereas enzymatic metabolism changes the molecule.

Mary Hare Bernheim described the enzyme system later called monoamine oxidase [41]. MAO-A participates in intracellular serotonin metabolism [25]. Cultured astrocytes can also take up serotonin through a carrier-mediated process [24], demonstrating a bounded multicellular contribution to local signal handling. This does not establish that all serotonin clearance is glial.

Other monoamine systems use distinct transporters and metabolic routes. Catecholamines can interact with catechol-O-methyltransferase in relevant tissues; that route must not be assigned indiscriminately to serotonin. “Signal termination” is a family of operations, not a single disposal step.

12.6 The signal ends by changing what can happen next

Receptor availability changes through synthesis, insertion, clustering, phosphorylation, desensitization, internalization, recycling, and degradation. Vesicle supply changes through refilling, membrane retrieval, reuse, and loss. Intracellular and neighboring-cell states can also change. The lifecycle therefore returns neither sender nor receiver to a guaranteed identical baseline.

A compact state update is

$$x_i(t + \Delta t) = F_i(x_i(t), c(\mathbf{r}, t), a_i(t), q_v(t), \theta_i(t), m_i(t), h_i[t - \tau, t]), \quad (9c)$$

where a_i is receptor availability, θ_i receptor configuration and coupling, m_i metabolic and intracellular state, and h_i recent history. The carrier-specific models below expand this compact form by specifying which variables persist at which scale.

The complete lifecycle is therefore:

```
precursor availability
-> synthesis
-> vesicular loading and storage
-> calcium-triggered release
-> diffusion and binding
-> receptor-dependent transduction
-> autoreceptor and local feedback
-> transporter uptake, neighboring-cell uptake, and metabolism
-> receptor and vesicle trafficking
-> return toward baseline or transition into a new receiver state
```

Vibration-assisted transport remains a source-gated cross-domain branch inherited from the earlier SET manuscript. It is not used as biological evidence here. It can strengthen SET only after the physical system, relevant temporal mode, transport observable, and boundary conditions are tied to a primary source.

13 Drugs and hormones expose the architecture by perturbing it

The ordinary question is simple: if two compounds reach the same named receptor, why can their effects still differ? The receptor is not an on-off label. A ligand can differ in affinity, efficacy, residence time, membrane

permeability, receptor conformation stabilized, pathway recruitment, and the cellular state in which it arrives. Drugs are therefore useful to SET not as a catalogue of effects but as controlled perturbations of the receiver-and-lifecycle architecture.

13.1 Receptor-directed perturbations

An agonist stabilizes receptor states capable of signaling in a defined system. A partial agonist produces a lower maximal response than a reference agonist in a specified assay. An antagonist reduces agonist access or response without necessarily activating the same route. An inverse agonist reduces measurable constitutive activity. These are operational definitions: the label can change with receptor reserve, assay, cell type, concentration, and downstream coupling.

Daniel Wacker and colleagues' LSD-bound serotonin-receptor structure supplies a concrete reason to include time: a ligand's residence and the receptor conformation it stabilizes can contribute to prolonged signaling [16]. David E. Nichols's pharmacological synthesis makes clear why "binds a serotonin receptor" is only the beginning of a mechanistic description [47].

13.2 Transporter and enzyme perturbations act at different lifecycle stages

An SSRI changes extracellular serotonin dynamics by inhibiting SERT-mediated reuptake [23]. A monoamine-oxidase inhibitor changes intracellular metabolic termination. These perturbations can interact, but they do not perform the same operation. SET predicts that interventions at different lifecycle stages can produce similar gross readouts while leaving different hidden states, recovery times, spatial ranges, and later sensitivities.

That prediction can be tested by matching an immediate readout, then measuring transmitter time course, receptor availability, second messengers, gene expression, and response to a later challenge. If the later state is identical once ordinary pharmacokinetic and receptor variables are controlled, the claimed SET distinction fails in that system.

13.3 Psychedelics test location, residence, efficacy, and pathway claims

Wacker and colleagues connect ligand residence and receptor structure to time-dependent signaling [16]. David Olson and colleagues reported that intracellular access to 5-HT_{2A} receptors contributed to measured plasticity outcomes in specified cultured-neuron and mouse experiments [43]. This makes receptor coordinate experimentally consequential rather than metaphorical.

The literature does not justify a universal claim that one G-protein-versus-arrestin bias separates psychedelic from non-psychedelic action. Alessandro Ippolito and colleagues found that the tested compounds in their matched system were not consistently separated by Gq versus beta-arrestin2 bias; lower overall efficacy better distinguished the non-psychedelic agonists in that assay [44]. SET should therefore model ligand identity, access, location, residence, efficacy, pathway recruitment, and receiver history as separate variables whose relevance must be measured.

13.4 Hormones demonstrate architectural generality, not interchangeability

Sutherland used hormone action to establish second-messenger logic [42]. Elwood Jensen's estrogen-receptor program established a different receiver architecture in which ligand access to an intracellular receptor changes transcriptional control [46]. These famous scientific landmarks add explanatory force because they identify distinct mechanisms:

```
ligand
-> receptor class and cellular coordinate
-> receiver-specific transduction
-> intracellular state change
-> feedback, persistence, and action
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The shared pattern does not make serotonin and estrogen interchangeable. Their synthesis, transport, receptor location, timescale, tissue distribution, feedback, and termination differ. The SET claim is architectural: functional meaning arises from the relation among signal, receiver, state, route, and history.

13.5 Pharmacological failure condition

The pharmacological extension fails if receptor configuration, signal-lifecycle stage, location, efficacy, and prior state add no reproducible predictive or interventional value beyond existing reduced descriptions. A famous name, a drug effect, or a receptor structure cannot rescue an under-specified model. Each historical citation must perform a defined causal job, and each proposed SET variable must earn its place through measurement.

14 Why timing can function as structure

A static parts list cannot say which route is open now.

A receptor can be present but desensitized. A voltage-sensitive channel can be present but unavailable at the current membrane state. A transcription factor can arrive without its necessary partner. A bacterium can enter or leave a competence window. An enzyme can be saturated. Feedback can reverse the effect of continued input.

Alan Turing's morphogenesis paper matters because it showed mathematically how temporal reaction and diffusion dynamics can generate spatial biological pattern [4]. Ilya Prigogine's work on nonequilibrium organization matters because living order is maintained through ongoing flows rather than static equilibrium. Naama Barkai and Stanislas Leibler later showed that a biochemical network can implement robust adaptation despite variation in many parameters, using bacterial chemotaxis as their central case [9]. Jon Kleinberg's burst-detection work supplies a narrower methodological analogy: temporal clustering can reveal a change of latent state rather than featureless noise [10].

SET's phrase **time is structure** means that timed state can function as effective control structure. It does not mean that duration is literally molecular anatomy. A route available at one phase, voltage, concentration, or feedback state may be unavailable at another while the molecular parts remain present.

15 From timed difference to cellular reconstruction

A cell receives partial evidence through a spatially distributed surface. Different receptors sample different chemical and physical variables. Their signals converge through membranes, second messengers, organelles, the cytoskeleton, transcriptional machinery, and neighboring cells. Coincident inputs can cross thresholds that separated inputs do not. Inhibition can close one route while leaving another available. Feedback changes the next response.

The plain-language operation is:

```
distributed partial signals
-> receiver-dependent transformation
-> comparison with maintained state
-> route-specific excitation or inhibition
-> changed cellular action and readiness
-> feedback from the consequence
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SET calls the proposed reconstruction of actionable state from distributed, time-structured cellular differences **Cellular Oscillating Tomography (COT)**. The changed cell is the registration of the result; no miniature inner observer is required.

The term **Phase Wave Differential (PWD)** is also earned from an operation rather than introduced as a synonym for phase coding. Let $(\phi_k(t))$ represent the timing or phase of input route (k), and let $(\widehat{\phi}_{ik}(t))$ represent the relation expected or maintained by receiver (i):

$$\Delta\phi_{ik}(t) = \text{wrap}\left(\phi_k(t) - \widehat{\phi}_{ik}(t)\right). \quad (10)$$

A PWD is a causally relevant, receiver-relative departure between an arriving time-structured event and the receiver's expected or maintained state. Its consequence depends on receptor tuning, inhibition, membrane state, and available pathways.

For an event (z) conditioned on receiver state (x_i), Shannon self-information is

$$\mathcal{I}(z \mid x_i) = -\log_2 P(z \mid x_i). \quad (11)$$

Common events have low surprisal under the model; uncommon events have higher surprisal. Rarity is not automatically biological meaning. A rare event can be noise or damage. SET adds a causal requirement: a receiver must discriminate the event and the event must change subsequent processing reproducibly.

16 How a transient response becomes a retained cellular difference

16.1 Begin with the ordinary problem

Imagine two cells that are presently similar by every measurement available to an investigator. Give both cells the same ligand at the same concentration and time. One responds strongly; the other responds weakly. The difference may reflect noise, an unmeasured present variable, damage, or a physical trace left by different prior histories.

The last possibility is the problem of cellular memory. It cannot be established by observing persistence alone. The investigator must control the histories, match the later input, identify the retained carrier, perturb that carrier, and show that erasing or blocking it changes the predicted response.

The relevant sequence is:

\\‘text controlled prior history -> retained physical difference -> matched later input -> history-dependent response -> carrier-specific blockade or erasure -> predicted loss of the response difference -> restoration or rescue where technically possible \\‘

This definition is deliberately stricter than saying that a cell has changed.

16.2 Retention, memory, learning, association, and heredity are different claims

Cellular state retention means that a physical variable persists after an initiating event. A calcium concentration, phosphorylation state, internalized receptor pool, metabolic condition, or chromatin mark can persist without necessarily changing a relevant later response.

Cellular memory means that a controlled history leaves an identifiable physical state that changes the distribution of responses to a matched later input. The retained state must outlast the initiating event and must be testable through intervention.

Cellular learning adds a declared criterion. Experience changes a retained state and thereby changes later discrimination, adaptation, prediction, survival, or task performance. A change can be maladaptive, so history dependence alone does not establish improvement.

Associative learning is stronger. It requires the effect to depend on a controlled relationship between events and to survive unpaired, exposure-only, sensitization, fatigue, damage, and motor-depletion controls.

Heredity asks whether the state crosses a division, reproductive, or transfer boundary through a specified carrier. Receptor trafficking is not mutation. Transcription is not genetic inheritance. Chromatin persistence is not automatically transgenerational inheritance.

These distinctions protect the strongest SET argument. SET can connect multiple forms of retained state without pretending they are the same physical process.

16.3 Memory requires a carrier and a clock

Represent the retained state of cell i as a multiscale carrier vector:

$$\mathbf{c}_i(t) = \left(c_i^{\text{elec}}, c_i^{\text{rec}}, c_i^{\text{met}}, c_i^{\text{prot}}, c_i^{\text{morph}}, c_i^{\text{chrom}}, c_i^{\text{gen}} \right) (t), \quad (12a)$$

where the components denote electrical or ionic state, receptor and trafficking state, metabolic state, protein or post-translational state, morphology, chromatin or regulatory state, and genetic or transferred state. No experiment must measure every component. The vector forces each claim to identify the proposed carrier instead of using the word memory as an unspecified substance.

A generic carrier update can be written as

$$\mathbf{c}_i(t + \Delta t) = \Phi_i(\mathbf{c}_i(t), u_i(t), x_i(t), E(t)) - \Lambda_i(\mathbf{c}_i(t)), \quad (12b)$$

where Φ_i contains reinforcement, modification, synthesis, trafficking, or structural stabilization and Λ_i contains decay, turnover, dilution, reversal, or active erasure. Francis Crick's molecular-turnover question earns its place here: a long-lived memory cannot be explained merely by naming a short-lived molecule [63]. The theory must specify how a pattern is renewed, copied, locally reinstated, or transferred while its components turn over.

The timescales overlap, but they must not be collapsed:

- milliseconds to seconds: voltage, ionic gradients, channel availability, and second messengers;
- seconds to minutes: phosphorylation, allosteric state, desensitization, and calcium-state history;
- minutes to hours: receptor methylation, trafficking, altered protein activity, and metabolic reprogramming;
- hours to days: translation, local protein synthesis, synaptic tagging, cytoskeletal and morphological change;
- days to cell generations: chromatin accessibility and stable regulatory state in bounded systems;
- generations or transfer events: selected sequence variation, mobile elements, plasmids, viruses, or symbiotic acquisition.

The carrier's lifetime sets the legitimate time horizon of the claim.

16.4 Receptor modification shows how history can change gain

Daniel Koshland's chemotaxis program provides a concrete answer to the question, "How can a cell's prior exposure change its later response without changing the ligand?" Spudich and Koshland showed that genetically similar bacterial cells can maintain different behavioral states [50]. Barkai and Leibler supplied a network architecture for robust adaptation [9], and Alon and colleagues tested robustness across large changes in protein abundance [48]. Levit and Stock then showed that receptor methylation can change stimulus-response gain in bacterial chemotaxis without being reduced to a change in ligand affinity [49].

The famous name matters because Koshland's work turns an abstract claim into a modifiable receiver mechanism:

\\'text prior ligand history -> receptor modification state -> changed coupling gain -> changed response to later ligand \\'

This is an established example of receptor-state adaptation. It is not evidence that every receptor modification is learning, that bacteria are conscious, or that receptor methylation is a DNA mutation.

Dennis Bray broadened the operational vocabulary by treating protein networks as cellular elements capable of amplification, integration, and state retention [51]. SET inherits that physical meaning of computation: a molecular network transforms conditions and changes future state. It does not require digital symbols or a miniature programmer inside the cell.

16.5 Timing can select gene programs without becoming a universal code

The next question is whether two inputs with similar aggregate dose can produce different outcomes because their timing differs. Work associated with Roger Tsien and Richard Lewis supplies a measured answer in calcium signaling. Dolmetsch, Xu, and Lewis showed that calcium-oscillation frequency affected the efficiency and specificity of gene expression in their preparation [52]. Li, Llopis, Whitney, Zlokarnik, and Tsien used controlled inositol-trisphosphate delivery to show that temporal organization could change transcriptional output even when aggregate input was constrained [53].

Galit Lahav and colleagues later showed that NF-kappaB and p53 dynamics can alter gene-expression trajectories or cell fate. Nelson and colleagues linked NF-kappaB oscillations to target-gene dynamics [54]. Purvis and colleagues showed that pulsed and sustained p53 activation could recruit different gene programs and outcomes [55].

These results support a bounded proposition:

\\‘text same pathway identity + different temporal organization -> different downstream integration -> different expression or fate trajectory \\’

They do not show that frequency alone contains meaning. Concentration, duration, localization, receptor state, partners, chromatin, metabolism, and noise remain causal. In SET, a timed difference becomes informative only relative to a receiver capable of discriminating it and only when a reproducible consequence follows.

This is where Claude Shannon’s information measure and SET’s PWD concept meet without becoming synonyms. Shannon quantifies how unexpected an event is under a conditional model. A PWD identifies a receiver-relative departure that changes a physical route. High surprisal without receiver discrimination may be noise; phase change without a downstream consequence may be irrelevant.

16.6 Neural memory shows why transient signaling must recruit stabilization

Donald Hebb made coordinated activity central to the problem of experience-dependent connection change [56]. Hebb’s rule family does not identify every molecular carrier, but it poses the right causal question: what changes when coordinated activity repeatedly recruits the same route?

Eric Kandel earns a larger role because his work connects short-lived signaling to long-lived synaptic change through gene regulation, protein synthesis, local synaptic mechanisms, and structural stabilization [60]. Bailey and Chen linked long-term habituation and sensitization in *Aplysia* to defined morphological differences [57]. Frey and Morris’s synaptic-tagging experiments showed how a local event can capture products needed for persistent potentiation [58]. Matsuzaki and colleagues related locally induced potentiation to enlargement of identified dendritic spines [59].

These findings establish neither one universal memory molecule nor a simple equation between morphology and remembered content. They establish a carrier chain:

\\‘text activity history -> local biochemical eligibility or tag -> gene expression or protein synthesis -> local capture and structural stabilization -> altered future synaptic response \\’

SET’s receptor and COT language adds a receiver-centered interpretation: the cell’s future discrimination depends partly on the physical arrangement produced by prior activity. A dendrite, receptor field, or synapse can constrain what will be received and amplified next. Demonstrating that constraint is weaker than proving a complete representational code, and the manuscript keeps those claims separate.

16.7 Self-reinstating protein states address part of Crick’s turnover problem

Kausik Si, Eric Kandel, and colleagues showed that a neuronal CPEB isoform contributes to local protein synthesis and stable maintenance of long-term facilitation in *Aplysia* [61]. Heinrich and Susan Lindquist demonstrated a self-perpetuating CPEB state in a bounded heterologous system [62].

The relevance is precise. A locally triggered molecular state can help maintain the machinery that renews a

synaptic modification. This supplies one answer class to Crick’s turnover problem: the carrier need not be one immortal molecule if an organization can regenerate its own state.

The boundary is equally important. A self-perpetuating protein state is not, by itself, the complete semantic content of a memory. It may be a maintenance mechanism, an eligibility condition, or one component in a larger distributed trace.

16.8 Chromatin can retain exposure history across longer intervals

Hathaway and colleagues engineered chromatin domains whose induced state persisted across cell generations long enough to measure competition between marking and turnover [64]. Saeed, Mihai Netea, Hendrik Stunnenberg, and colleagues linked trained innate immunity to durable metabolic, transcriptional, and epigenomic reprogramming in defined monocyte/macrophage preparations [65]. De Laval and colleagues identified a C/EBPbeta-dependent chromatin-accessibility program in hematopoietic stem and progenitor cells associated with altered secondary response [66].

These studies make a stronger persistence claim than transient second-messenger carryover, but a narrower claim than unrestricted heredity. They show that prior exposure can leave regulatory accessibility that changes a later response within specified preparations and boundaries.

Conrad Waddington’s landscape becomes mechanistically useful here. A landscape is not merely a picture of possible phenotypes. Chromatin accessibility, transcription-factor availability, metabolism, and feedback can change which routes are physically easier to enter next. SET therefore uses Waddington as a path-dependence constraint, not as proof of a new molecule.

16.9 Non-neural learning requires stronger controls than evocative behavior

Boisseau, Vogel, and Dussutour reported habituation-like behavior in the multinucleate slime mould *Physarum polycephalum*, including reduced avoidance after repeated exposure and recovery after rest [67]. The result is relevant because it tests history-dependent behavior in a non-neural system.

It does not license the claim that every cell thinks. Habituation must be separated from fatigue, receptor saturation, damage, motor depletion, and generalized response loss. Associative learning requires additional paired-versus-unpaired controls. Cellular memory requires an identifiable retained carrier.

SET gains credibility by preserving these distinctions. The theory predicts where stronger evidence should be sought; it does not award every persistent response the strongest label.

16.10 The joined SET memory operator

Let h_A and h_B denote controlled prior histories, u a matched later input, x_{obs} matched measured present covariates, and $Y_{t+\tau}$ the later response. A history-sensitive system satisfies

$$P(Y_{t+\tau} \mid u, x_{\text{obs}}, h_A) \neq P(Y_{t+\tau} \mid u, x_{\text{obs}}, h_B). \quad (12c)$$

A bounded history-separation statistic is

$$M_\tau(h_A, h_B; u, x_{\text{obs}}) = D_{\text{JS}}(P_A(Y_{t+\tau}) \parallel P_B(Y_{t+\tau})), \quad (12d)$$

where D_{JS} is the symmetric, finite Jensen-Shannon divergence between response distributions under the two histories. A positive M_τ shows predictive history dependence under the measured model. It does not yet identify a carrier.

For a candidate carrier c , define an intervention score

$$E_c(\tau) = M_\tau^{\text{intact}} - M_\tau^{\text{blocked or erased}}. \quad (12e)$$

A credible carrier should predict a positive, reproducible E_c , and a selective rescue should restore the response difference where technically possible. The study must also test whether a better measured present-state model eliminates the apparent history effect. If adding the carrier to x_{obs} restores state sufficiency, the result identifies missing state rather than a failure of Markov mathematics.

The resulting SET sequence is therefore:

\\‘text distributed signal and receiver state -> receiver-relative PWD or other discriminated event -> electrical, biochemical, metabolic, or transcriptional consequence -> stabilization in a named physical carrier -> changed receptor availability, pathway gain, morphology, or regulatory accessibility -> altered response to a matched later input -> reinforcement, decay, erasure, selection, or inheritance at the carrier’s actual boundary \\‘

COT names the proposed reconstruction of actionable cellular state from distributed, time-structured differences. PWD names a receiver-relative departure capable of changing a route. Neither term automatically denotes long-term memory. Memory begins only when a consequence is stabilized in a carrier and later changes response.

16.11 Discriminating experiments

A cellular-memory experiment should include:

1. at least two controlled prior histories;
2. a matched later input;
3. sufficiently rich present-state measurements to challenge the hidden-history explanation;
4. a declared candidate carrier and timescale;
5. carrier blockade or erasure;
6. selective rescue where possible;
7. held-out prediction of the later response;
8. fatigue, damage, depletion, sensitization, and nonspecific-state controls;
9. a test of decay and renewal;
10. an explicit inheritance boundary.

Failure condition: the response difference disappears when present state is adequately measured, cannot be tied to a retained carrier, survives no carrier-specific intervention, or reduces to nonspecific damage, depletion, or selection among pre-existing states.

17 Evolution is descent, variation, selection, and sometimes transfer

17.1 Begin with the ordinary evolutionary problem

Imagine two lineages facing the same environmental challenge. One lineage must assemble a useful function through lineage-local mutation, recombination, duplication, and regulatory change. The other encounters a compatible module that evolved elsewhere.

The second lineage has not escaped evolution. A donor must exist. The module must be carried, transferred, admitted, expressed, retained, and tolerated. Its effects remain subject to selection and drift. Yet the set of available transitions has changed. A route exists that a lineage-only model did not include.

This produces the evolutionary-network problem:

When does moving a preassembled biological module change evolutionary reachability or waiting time, and when do incompatibility, cost, ecology, loss, or selection erase the apparent shortcut?

The answer requires more than saying that evolution is a network. It requires a physical carrier, a transport route, a receiver, rates, costs, and observable outcomes.

17.2 The modern baseline is not “pure randomness”

Charles Darwin earns the first place in this section because descent with modification and natural selection make differential survival and reproduction central to evolutionary explanation [68]. Darwin did not possess genes, population genetics, mobile elements, plasmids, or molecular phylogenies. SET therefore inherits Darwin’s selection logic without pretending that nineteenth-century mechanism exhausts the sources or routes of variation.

Ronald Fisher, J. B. S. Haldane, and Sewall Wright made evolutionary change mathematical. Fisher joined Mendelian inheritance to selection [69]. Haldane modeled selection and mutation as explicit rates [70]. Wright added population structure, drift, and the adaptive-landscape picture already used in SET [3]. Their disagreements matter less here than their shared constraint: evolutionary claims must specify population state, variation, transition, inheritance, and reproductive consequence.

Motoo Kimura earns a separate role because neutral molecular evolution blocks an easy overstatement [71]. Not every retained molecular difference is an adaptation, and not every fixed change was driven by positive selection. SET must allow neutral and nearly neutral changes, genetic drift, linkage, demographic history, and background selection alongside adaptive module reuse.

The baseline is therefore:

\\‘text variation from several physical mechanisms -> inheritance or loss -> population- and ecology-dependent consequences -> selection, drift, linkage, and demographic change -> altered frequencies and future possibilities \\‘

SET does not replace this sequence. It adds receiver state and non-lineage-local transport where the evidence requires them.

17.3 Duplication and regulation create new receiver possibilities

Susumu Ohno's *Evolution by Gene Duplication* matters because duplication creates a second copy whose future can differ from the first [72]. François Jacob's "evolutionary tinkering" matters because evolution often modifies and recombines available parts rather than beginning from an unconstrained design [74]. Mary-Claire King and Allan Wilson matter because regulatory change can contribute to phenotype even when many protein sequences remain similar [75].

These names mark three different operators:

\\‘text duplication -> additional copy and dosage

divergence -> loss, partition, altered expression, or altered function

regulatory change -> same or related component deployed in a new amount, place, time, or context \\‘

Lynch and Conery's duplicate-gene analysis constrains the inspiring version of Ohno's story: many duplicates are lost, and retained copies face population-genetic and functional constraints [73]. Duplication is an opportunity distribution, not guaranteed novelty.

SET's receptor ecosystem depends on exactly this distinction. A new receptor subtype can differ through sequence, ligand sensitivity, coupling, localization, expression, regulation, or partner availability. A duplicated receptor that is never expressed, cannot couple, or imposes excessive cost adds no effective channel.

17.4 The serotonin receptor example has an evolutionary history

Peroutka and Howell used phylogenetic comparisons to reconstruct an early history of G-protein-coupled serotonin receptors [76]. Their molecular-clock estimates reflect the sequences and models available in 1994 and should not be treated as timeless dates. Their central relevance survives: the present 5-HT receptor repertoire is an evolved family, not a set of labels created at once.

The receptor atlas can therefore be read as an evolutionary outcome:

\\‘text ancestral receptor and signaling machinery -> duplication or genomic expansion -> divergence and differential loss -> altered coupling, ligand response, expression, and localization -> cell- and tissue-specific receiver repertoires -> selection, drift, and developmental constraint \\‘

The ionotropic 5-HT₃ family and the metabotropic 5-HT GPCR families also warn against a single simple tree for "serotonin reception" as one operation. Similar ligand responsiveness can be implemented through deeply different receptor architectures. The evolutionary analysis must distinguish homology, analogy, functional convergence, and pathway recruitment.

17.5 Three classic transfer routes move different carriers

The word transfer becomes scientific only when its route is named.

Transformation. Oswald Avery, Colin MacLeod, and Maclyn McCarty showed that a DNA fraction could carry a stable transforming property between pneumococcal cells [77]. In modern terms, extracellular DNA must survive, encounter a competent receiver, enter, evade degradation, and integrate or persist.

Conjugation. Joshua Lederberg and Edward Tatum established bacterial genetic exchange and

recombination in *Escherichia coli* [78]. Later work identified plasmids and cell-contact machinery. A conjugative element must encounter a compatible host, execute transfer, replicate or integrate, and survive cost and segregational loss.

Transduction. Norton Zinder and Joshua Lederberg established phage-mediated genetic exchange in *Salmonella* [79]. A phage or phage-derived carrier must match host range, package relevant material, deliver it, and permit integration or expression. Large-scale reconstruction by Popa, Landan, and Tal Dagan shows why transduction edges have limited phylogenetic range rather than connecting every genome equally [86].

Thomas and Nielsen's mechanism-and-barrier map is useful because it names the filters between possible and successful transfer [80]. Transformation, conjugation, and transduction are not interchangeable metaphors. They expose different contact rules, carriers, entry barriers, host ranges, and persistence conditions.

17.6 Mobile elements and symbiosis change different boundaries

Barbara McClintock's mobile controlling elements provide a physical mechanism for movement and regulation within a genome [26]. A transposon can relocate or copy sequence without moving between cells. That is a topology change inside a genome, not automatically horizontal gene transfer.

Lynn Margulis's endosymbiotic argument, published under Lynn Sagan, supplies a much larger combination operator [27]. An entire living system can enter a persistent relationship that reorganizes cellular architecture over evolutionary time. Endosymbiosis is not merely a large plasmid transfer; it changes organismal and cellular boundaries.

François Jacob's tinkering, McClintock's transposition, Ohno's duplication, and Margulis's symbiosis all involve reuse, but the carriers and transitions differ. SET keeps the family resemblance while refusing to flatten the mechanisms.

17.7 A biological packet must earn the name

SET calls a transferable unit a **packet** only when eight properties are specified:

1. a physically bounded carrier;
2. identifiable coding, regulatory, catalytic, or structural content;
3. a transport route;
4. receiver-side entry compatibility;
5. an expression, integration, replication, or assembly path;
6. a persistence mechanism;
7. an observable consequence;
8. a selectable or drift-sensitive population outcome.

A plasmid can satisfy all eight in one system and fail at entry, expression, retention, or benefit in another. A transposon may supply content and mobility inside one genome but lack an intercellular route. A phage can provide transport but face narrow host range. An organelle or symbiont brings a larger integrated system with different maintenance requirements.

The word packet therefore names a governed comparison class. It does not claim that DNA uses Internet protocols or that all carriers have the same architecture.

17.8 Receiver compatibility is part of evolutionary information

Let donor d , receiver i , and module m interact. Define a successful-transfer rate

$$\lambda_T(d, i, m, t) = c_{di}(t) \tau_m(t) s_m(t) \kappa_{im}(t) \epsilon_{im}(t) \rho_{im}(t), \quad (13a)$$

where:

- c_{di} is donor-receiver contact rate;
- τ_m is transfer probability per qualifying contact;
- s_m is carrier survival or delivery probability;
- κ_{im} is entry, replication, integration, or assembly compatibility;
- ϵ_{im} is probability of functional expression in the receiver;
- ρ_{im} is retention probability over the declared observation interval.

The factorization is an experimental bookkeeping model, not a claim that the variables are always independent. A zero or near-zero factor blocks the route. Receiver compatibility is therefore as central at the evolutionary scale as receptor configuration is at the cellular scale.

Loftie-Eaton and colleagues showed that host evolution can increase general permissiveness to resistance plasmids [84]. Bottery, Wood, and Brockhurst showed that plasmid and chromosome can coadapt, reducing cost while potentially making the combination less portable [85]. Receiver state changes not only the present effect of a signal but also the future accessibility of transferred biological content.

17.9 Trees remain useful, but one tree may not contain every history

A species tree represents descent. A gene tree represents the history inferred for a particular gene or gene family. Recombination, hybridization, duplication, loss, incomplete lineage sorting, and transfer can make those histories differ.

Carl Woese's communal-ancestor proposal gives early-life innovation sharing a specific historical role [28]. W. Ford Doolittle explains why genes from multiple sources complicate a universal organismal tree [29]. Kunin and colleagues explicitly reconstructed a microbial "net of life" [30]. Dagan and Martin modeled prokaryotic genomes as modular networks containing both vertical and lateral contributions [89].

Their names mark four different claims: an early communal regime, limits on one universal tree, a reconstructed gene-sharing network, and a modular quantitative network model. None implies that vertical descent disappears or that every biological history is equally reticulate.

The correct representation can be multilayered:

$$\mathcal{G} = (V, E_{\text{vertical}}, E_{\text{duplication}}, E_{\text{recombination}}, E_{\text{transposition}}, E_{\text{transfer}}, E_{\text{symbiosis}}). \quad (13b)$$

Each edge type requires its own evidence and carrier.

17.10 Reachability, shortest path, and waiting time are not the same quantity

Let $\mathcal{G}_L = (V, E_L)$ contain lineage-local transitions, and let $\mathcal{G}_{L+T} = (V, E_L \cup E_T)$ add empirically allowed transfer edges. For any starting state x ,

$$\text{Reach}_{\mathcal{G}_L}(x) \subseteq \text{Reach}_{\mathcal{G}_{L+T}}(x). \quad (13c)$$

For any target y already reachable in $\mathcal{G}_L$, unweighted shortest-path distance satisfies

$$d_{\mathcal{G}_{L+T}}(x, y) \leq d_{\mathcal{G}_L}(x, y). \quad (13d)$$

These graph statements are elementary and appropriate for Lean verification. They do not imply that a population will traverse the new edge, that transfer is probable, or that the acquired state will persist.

Under a deliberately narrow independent-hazard model, suppose local assembly of a target module occurs at rate $\lambda_L > 0$, and successful transfer routes add total rate

$$\lambda_T = \sum_{d,m} \lambda_T(d, i, m). \quad (13e)$$

If the first local or transfer arrival is modeled as the minimum of independent exponential clocks, then

$$\mathbb{E}[T_{\text{first}}] = \frac{1}{\lambda_L + \lambda_T} \leq \frac{1}{\lambda_L}. \quad (13f)$$

This is the clean mathematical core of SET's original **wormhole effect**: an added successful route can reduce first-arrival waiting time under explicit assumptions. It is not a universal law of evolution. If transfer diverts resources, changes local mutation supply, introduces conflict, or alters the transition kernel, expected time can increase.

17.11 Arrival is not persistence or fixation

After acquisition, let the net context-dependent effect of module m in receiver i be

$$s_{im}(E) = b_{im}(E) - c_{im}^{\text{expression}} - c_{im}^{\text{replication}} - c_{im}^{\text{conflict}} - c_{im}^{\text{ecology}}, \quad (13g)$$

where benefits and costs depend on environment E . This compact expression does not replace population genetics. It makes explicit why one module can help in one receiver and harm in another.

Experimental plasmid studies provide both favorable and adverse cases. Ellie Harrison and colleagues showed that rapid compensatory evolution can promote the survival of conjugative plasmids in experimentally evolved bacteria [31]. San Millan and colleagues showed that positive selection and compensatory adaptation can jointly stabilize a costly plasmid [81], and that multicopy plasmids can accelerate resistance evolution through mutation supply and dosage in a defined system [82]. Hall and colleagues found that positive selection inhibited some gene mobilization and interspecific transfer in a soil community [83]. Wein and colleagues showed that plasmid stability can emerge through changes in transcription-replication conflict without selection for the focal resistance function [87]. Nguyen and colleagues demonstrated that repeated HGT of an ancestral allele altered the probability of reverse evolution in *Helicobacter pylori* [88].

The lesson is not “transfer accelerates evolution.” The defensible result is conditional:

\\‘text carrier availability x contact and transfer x receiver compatibility x expression and retention x ecological consequence x selection and drift -> observed evolutionary outcome \\‘

Any factor can reverse the conclusion.

17.12 Externalized biological memory is a bounded analogy

A transferable module embodies the history of its prior assembly. A recipient can acquire that organized content without repeating every event in the donor lineage. SET calls this **externalized biological memory** only in the bounded sense that a persistent module assembled elsewhere can alter the recipient’s future transition options.

The analogy does not imply recollection, experience, or consciousness. Its measurable content is provenance, structure, function, transfer route, compatibility, and consequence. Independent reconstruction must be distinguished from transfer through sequence, synten, carrier, donor-recipient, and phylogenetic evidence.

17.13 Discriminating evolutionary experiments

A module-reuse experiment should compare:

1. a lineage-local assembly condition;
2. a transfer-enabled condition with known donor and carrier;
3. a transfer-blocked control;
4. compatible and incompatible receivers;
5. expression-competent and expression-disabled modules;
6. cost-matched neutral cargo;
7. retention and segregational-loss trajectories;
8. positive, negative, and absent selection;
9. measured mutation supply and population size;
10. drift-sensitive replicate populations;
11. provenance sufficient to distinguish import from independent reconstruction;
12. held-out prediction of acquisition, persistence, and population outcome.

Failure condition: adding transfer variables does not improve prediction beyond mutation, recombination, population structure, ecology, and selection; the claimed carrier or route is absent; imported content cannot be distinguished from independent reconstruction; or the apparent shortcut vanishes after compatibility, cost, and loss are modeled.

18 The joined SET mechanism

The receptor atlas changes SET’s central causal sequence from a general analogy into a measurable architecture:

environmental or intercellular condition
 -> ligand identity, concentration, sequence, and timing
 -> receptor class, subtype, configuration, and coordinate
 -> ionotropic or metabotropic transformation

- > membrane, biochemical, metabolic, and rhythmic state change
- > feedback, trafficking, action, competence, or structural adaptation
- > stabilization in a named electrical, receptor, metabolic, protein, morphological, chromatin, genetic
- > altered response to a matched later input
- > differential persistence, survival, and reproduction
- > mutation, recombination, duplication, regulatory change, or module assembly
- > vertical inheritance, transposition, transformation, conjugation, transduction, or symbiotic combina
- > receiver-specific entry, expression, retention, cost, selection, and drift
- > changed receptor and regulatory possibilities in a recipient
- > new discrimination and action

This is more specific than saying “cells compute” or “evolution processes information.” It identifies receiver, input, transformation, state update, persistence mechanism, transport route, and observable consequence.

A provisional population transition model is

$$P(X_{t+1} \mid X_t, H_t, \mathcal{G}_t, E_t), \quad (14)$$

where (X_t) is measured population state, (H_t) is hidden history carried by cells and environments, ($\mathcal{G}_t$) is current lineage-and-transfer topology, and (E_t) is ecological condition. A memoryless, lineage-only approximation omits (H_t) and ($E_{\{\mathrm{transfer}\}}$). SET predicts that those omissions materially degrade prediction in specified regimes.

The theory is not “randomness is false.” Transition probabilities are conditioned by state and transition graphs can contain module-reuse routes. Stochastic variation can remain fundamental or practically irreducible while being shaped by receiver state, feedback, ecology, selection, and topology.

19 What SET inherits and what it proposes

19.1 Established scientific background

- Receptor subtype and cellular context influence signal outcome.
- Ionotropic and metabotropic serotonin receptors perform different operations.
- Gi/o, Gq/11, and Gs supply distinct principal transduction routes.
- Receptor editing, conformation, localization, trafficking, and feedback alter response.
- Transmitters are synthesized, stored, released, diffused, transported, and metabolized through distinguishable mechanisms.
- Vesicle recycling, receptor recycling, transmitter reuptake, and enzymatic degradation are distinct operations.
- Drugs can perturb receptor efficacy, access, residence, transport, metabolism, and feedback at different lifecycle stages.
- Hormones demonstrate other receptor architectures without being interchangeable with neurotransmitters.
- Cells retain state across multiple timescales.
- Receptor modification can alter response gain without changing ligand identity.
- Temporally organized calcium, NF-kappaB, and p53 dynamics can change transcriptional trajectories or fate in bounded systems.
- Neural plasticity can recruit gene expression, local protein synthesis, synaptic tagging, and morphological stabilization.

- Chromatin and regulatory state can preserve exposure history across cell generations in specified preparations.
- Gene duplication, differential loss, and regulatory change diversify receptor and control architectures.
- Transformation, conjugation, and transduction move different carriers through different routes and barriers.
- Mobile elements, symbiosis, plasmids, and horizontal transfer complicate a single lineage tree.
- Transfer success depends on contact, entry, expression, retention, cost, ecology, selection, and drift.
- Natural selection acts on the consequences of these mechanisms.

19.2 SET synthesis

- Treat the receptor ecosystem as a configurable input-and-control surface connecting signal discrimination to persistent adaptation.
- Model timed receiver state as effective routing structure.
- Represent the full signal lifecycle so synthesis, release, diffusion, reception, feedback, uptake, metabolism, and reset cannot be collapsed into one event.
- Use pharmacological interventions as stage-specific probes of hidden receiver state rather than as unstructured lists of drug effects.
- Join COT/PWD receiver-relative differences to longer-lived cellular state without pretending every difference becomes memory or heredity.
- Require every memory claim to identify its carrier, clock, erasure intervention, and inheritance boundary.
- Use matched-input history separation to test whether an apparently identical present state omits a causally retained variable.
- Model evolutionary history as a typed multilayer graph containing vertical, duplication, recombination, transposition, transfer, and symbiosis edges.
- Treat transfer as module reuse subject to donor contact, carrier survival, receiver entry, expression, retention, cost, ecology, selection, and drift.
- Separate reachability, shortest path, transition probability, expected waiting time, persistence, and fixation.
- Use packetized evolution only when the carrier, content, route, receiver, expression, persistence, consequence, and population outcome are specified.

19.3 Stronger SET hypotheses requiring tests

- Receptor-ecosystem state improves prediction across cell classes beyond ligand-only and standard reduced models.
- Oscillatory phase or timed state controls pathway accessibility after total dose is matched.
- Receptor-state histories create persistent biases large enough to change action, survival, competence, or transfer statistics.
- A multiscale carrier model predicts when transient electrical or biochemical differences will recruit protein, morphological, chromatin, genetic, or transferred persistence.
- Carrier-specific erasure and rescue distinguish SET memory claims from unmeasured-state, damage, depletion, and selection explanations.
- Compatible module transfer reduces expected evolutionary waiting time only in defined regimes where added successful-arrival rate outweighs diversion, conflict, and cost.
- Receiver permissiveness and compensatory adaptation predict which transferred modules persist and which become receiver-specific dependencies.
- A compositional rule can connect receptor discrimination, cellular state update, persistence, and reticulate evolution without conflating their carriers.

19.4 Alternative explanations SET must beat

SET does not earn value merely by placing established mechanisms beside one another. It must outperform, or reveal a missing interaction among, strong reduced explanations that already exist.

Langley's receptor distinction and modern IUPHAR pharmacology may explain an outcome through receptor identity, location, efficacy, and coupling without requiring the full SET architecture [1,11-17]. Barkai and Leibler's robustness analysis, Alon and colleagues' chemotaxis experiments, and Levit and Stock's receptor-methylation work may explain adaptation and reset through an identified biochemical feedback circuit [9,48,49]. Dolmetsch and Lewis, Tsien and colleagues, Nelson and colleagues, and Lahav's group may explain timing-dependent transcription or fate through pathway-specific dynamics [52-55]. Hebb, Kandel, synaptic-tagging, morphology, CPEB, chromatin, and trained-immunity mechanisms may explain persistence through carrier-specific models without invoking one general cellular-learning operator [56-66]. Darwinian population genetics, neutral evolution, duplication, regulatory change, mobile elements, HGT, plasmid ecology, and reticulate phylogeny may explain evolutionary outcomes without SET's packet vocabulary [68-89]. A Markov model can also absorb relevant history by enlarging its state rather than remaining memoryless at an impoverished observational level [90].

The discriminating burden is therefore compositional. SET must show that a declared combination of receiver configuration, time-structured input, retained carrier state, signal termination, and transport topology predicts a held-out consequence better than these reduced accounts, or that an intervention on one layer changes another layer in the direction specified by the joined model. If the same result is fully explained by receptor identity alone, ordinary adaptation, a single carrier, standard population genetics, or an appropriately augmented state model, the broader SET claim has not won that test.

20 Mathematical and formal program

20.1 What the executable model can and cannot establish

The executable model asks whether the manuscript's claims have distinct mathematical consequences. It does not fit biological data and it does not estimate a parameter for any organism. Synthetic variables are useful here because the generating assumptions are known: if an analysis cannot recover a receiver effect that was deliberately placed in the generator, cannot distinguish equal-dose temporal profiles in a timing-sensitive dynamical system, or cannot separate arrival from persistence, then the proposed discriminator is inadequate before biology is even attempted.

The model uses the frozen pseudorandom seed 20,260,809. Nine automated tests check prediction, dose matching, temporal sensitivity, carrier erasure and rescue, analytic-versus-simulated waiting time, persistence ordering, graph monotonicity, an adverse hitting-time counterexample, compatibility sensitivity, and output parsing. Every result is written to inspectable JSON or CSV, and the four main result families are rendered in the executable-model figure.

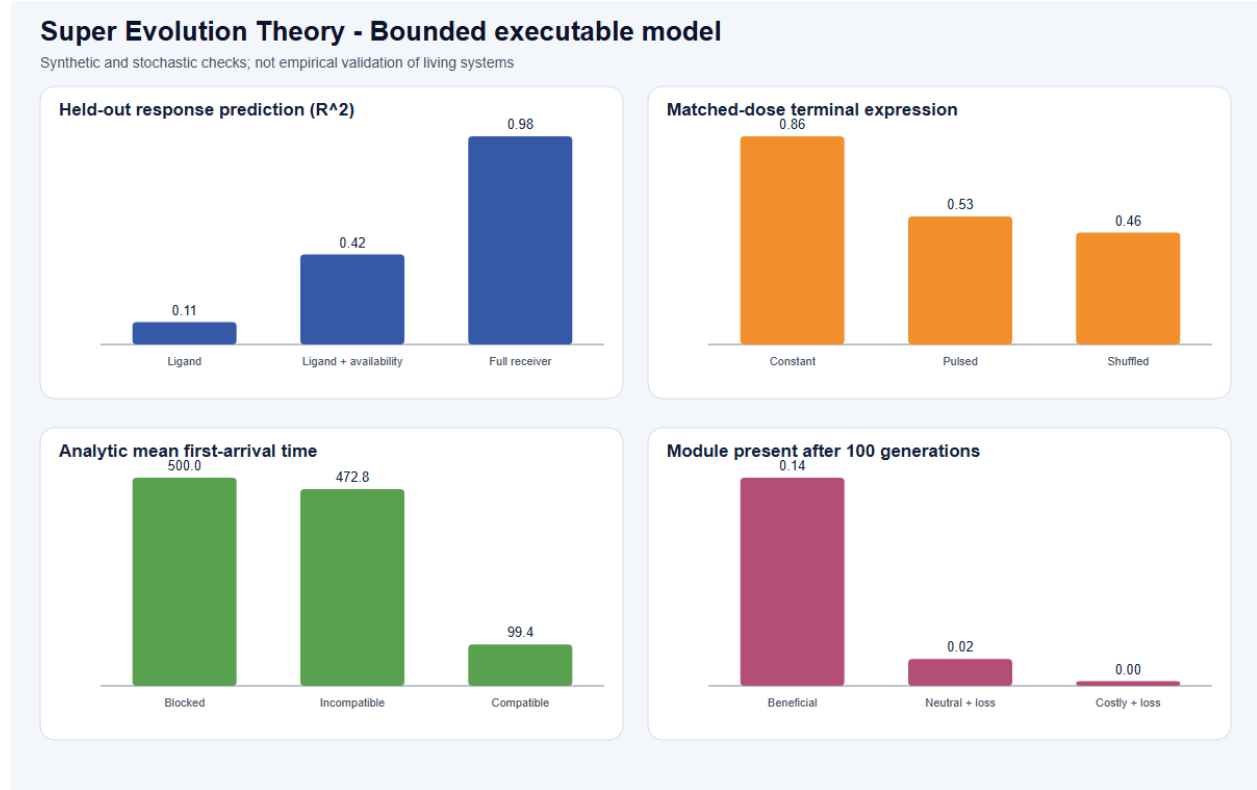


Figure 2: Bounded executable checks for receiver state, temporal organization, compatible transfer, and post-acquisition persistence. These are synthetic model consequences, not measurements of living systems.

20.2 Receiver state as a predictive variable

Claude Shannon's name matters here because information is conditional: a ligand becomes informative about a consequence only relative to the receiver variables that constrain the possible outcomes. The synthetic response generator therefore makes ligand concentration only one term in the operation:

$$Y = A \frac{L}{K + L} C (1 + \eta \cos \phi) + \omega M + \epsilon, \quad (14a)$$

where L is ligand concentration, A is receptor availability, C is coupling gain, ϕ is a phase-like receiver-state variable, M is retained history, and ϵ is noise. Three linear models are trained on 70% of 6,000 cases and tested on the held-out 1,800: ligand only; ligand plus availability; and the full receiver-state feature set, including interaction terms.

Because the generator is receiver dependent, the experiment is a recovery and identifiability test rather than evidence from nature. Ligand alone produces held-out $R^2 = 0.106$ and RMSE 0.251. Adding availability raises R^2 to 0.425. The full receiver-state model reaches $R^2 = 0.983$ and RMSE 0.0348, a gain of 0.877 in held-out R^2 over ligand alone. The scientific next step is not to celebrate a synthetic fit; it is to ask whether measured receptor, coupling, phase, and history variables produce a comparable out-of-sample gain in real perturbation data.

20.3 Equal dose is not equal temporal meaning

Alan Turing's reaction-diffusion work earns a place in SET because it showed that temporal and local interactions can generate spatial organization; Robert Tsien, Richard Lewis, Nelson, and Lahav later made timing-dependent gene regulation experimentally concrete. The executable model asks a narrower question: can three inputs with exactly the same integrated dose produce different downstream trajectories in a nonlinear receiver?

The model evolves a calcium-like state $c(t)$ and an expression-like state $e(t)$:

$$\frac{dc}{dt} = -\frac{c}{\tau_c} + k_c u(t), \quad \frac{de}{dt} = \frac{c^n}{K_c^n + c^n} - \frac{e}{\tau_e}. \quad (14b)$$

Constant, pulsed, and temporally shuffled inputs are each rescaled to integrated dose 2.0. Their terminal expression values are nevertheless 0.864, 0.532, and 0.464, respectively. The maximum dose difference is zero to numerical precision. This result establishes only that the stated nonlinear dynamical system distinguishes timing after dose matching. A biological SET test must measure the relevant state variables and must defeat alternatives such as saturation, depletion, damage, or a hidden dose mismatch.

20.4 Retained state, erasure, and rescue

Francis Crick's molecular-turnover question and Eric Kandel's signaling-to-structure program matter because a transient response is not yet a memory. The model therefore gives history a named bounded carrier:

$$M_{t+1} = \text{clip}[M_t + \alpha u_t(1 - M_t) - \delta M_t, 0, 1]. \quad (14c)$$

Two exposure histories create carrier values 0.599 and 0.261. A matched later probe then produces a response gap of 0.180. Setting the carrier to zero eliminates the gap; restoring the exact carrier states restores the exact gap. That perfect rescue is expected because the intervention is defined on the synthetic causal variable. Its purpose is to specify the burden for experiments: identify a carrier, predict the direction of its erasure, and restore the response selectively. If an intervention merely injures the cell or resets many variables at once, it has not passed the SET memory test.

20.5 Compatible transfer and first arrival

The transfer experiment operationalizes Equation (13a). A successful-transfer rate is the product of contact, transfer per contact, carrier survival, receiver compatibility, expression, and retention. A local innovation clock with rate $\lambda_L = 0.002$ competes with a transfer clock. Under the independent exponential-hazard assumption,

$$T_{\text{first}} = \min(T_L, T_T), \quad \mathbb{E}[T_{\text{first}}] = \frac{1}{\lambda_L + \lambda_T}. \quad (14d)$$

With transfer blocked, the analytic mean wait is 500. A nominal transfer route with receiver compatibility 0.01 barely changes it, yielding 472.77. With compatibility 0.70, the successful-transfer rate becomes 0.008064, the analytic mean falls to 99.36, and the modeled reduction is 80.13%. Monte Carlo estimates

from 100,000 samples remain within two percent of the analytic means. This is the bounded wormhole result: successful additional hazard lowers first-arrival time under these assumptions. It says nothing by itself about persistence, fixation, net fitness, or whether the independence assumption holds.

20.6 Wright-Fisher persistence after acquisition

Ronald Fisher and Sewall Wright earn their names here because acquisition is not evolutionary success. The model starts each haploid population of $N = 200$ with one acquired copy and evolves selection, retention, and binomial sampling:

$$p_t^* = \frac{p_t(1+s)}{1+sp_t}, \quad \tilde{p}_t = rp_t^*, \quad K_{t+1} \sim \text{Binomial}(N, \tilde{p}_t). \quad (14e)$$

Across 4,000 replicates and 100 generations, the beneficial compatible case ($s = 0.08, r = 0.999$) remains present in 13.78% of populations and fixes in 3.10%. The neutral case with loss ($s = 0, r = 0.995$) remains present in 1.78%, while the costly case ($s = -0.03, r = 0.995$) remains present in 0.30%. These values belong only to the stated toy population. Their importance is categorical: a route can accelerate arrival while drift, cost, or loss still removes the module.

20.7 Reachability, shortest paths, and stochastic time

Andrey Markov's study of dependent sequences matters because the transition-kernel tradition that bears his name provides a precise baseline against which hidden history and altered routing can be tested [90]. SET's criticism is therefore not that Markov models are incapable of history. A process can often be made Markovian by enlarging the state until the relevant history is represented. The test is whether a reduced observable state has omitted receptor history, cellular structure, ecology, or transport topology that measurably improves prediction. In a finite directed graph, adding edges cannot remove an already reachable node and cannot lengthen an existing unweighted shortest path. The executable example expands the reachable set and shortens the path to the target from four edges to two.

Expected hitting time answers a different question. Richard Bellman's dynamic-programming principle matters here because it turns a multistep future into a local recurrence over the next possible states [91]. For target T , that first-step recurrence is

$$h_T = 0, \quad h_i = 1 + \sum_j P_{ij} h_j \quad (i \neq T). \quad (14f)$$

In the counterexample, the baseline kernel reaches the target in an expected 2.0 steps. A modified kernel adds access to another state but assigns substantial probability to a sticky diversion; expected hitting time rises to approximately 2,600 steps even though the graph has more edges. The lesson is not that additional transfer usually slows evolution. It is that topology, transition probability, compatibility, and cost cannot be collapsed into one word such as "connectedness."

20.8 Reproducibility and formal-proof boundary

The accompanying reproducibility package contains the executable source, automated tests, frozen result set, compatibility sweep, and the data underlying Figure 1. All nine tests pass under the declared frozen seed.

The accompanying formal package pins Lean 4.26.0 and Mathlib 4.26.0, records the exact Mathlib commit, builds successfully, and passes separate axiom and placeholder audits. Machine-checked proofs establish consequences of stated models. They do not establish that those models describe living cells; experiments must do that.

20.9 What Lean proves

The graph kernel defines a finite-length path directly. It proves that if every old edge remains present, every old path remains present with the same length. Reachability therefore cannot shrink, and any previously witnessed path-length upper bound remains available. This is the exact machine-checked result. A theorem stated directly in terms of a canonical minimum-distance function remains open; the manuscript therefore does not label that stronger formulation as formally certified yet.

The transfer kernel defines the successful-transfer rate as the product of contact, transfer per contact, carrier survival, receiver compatibility, expression, and retention. Lean proves that the product is nonnegative when every factor is nonnegative. It then proves

$$\frac{1}{\lambda_L + \lambda_T} \leq \frac{1}{\lambda_L} \quad (15a)$$

for $\lambda_L > 0$ and $\lambda_T \geq 0$, with strict inequality when $\lambda_T > 0$. A composed theorem inserts the six-factor transfer rate into the same result. This certifies the bounded independent-hazard claim and nothing stronger.

The carrier kernel proves that a positive probe and nonnegative half-saturation produce a positive response denominator. It proves that equal erased carrier states yield zero response difference, that exact restoration reproduces the exact modeled difference, and that the difference factors as

$$R(M_A) - R(M_B) = \frac{p}{K + p} g(M_A - M_B). \quad (15b)$$

The factorization makes the intervention logic explicit: in this model, the response difference vanishes if the carrier difference vanishes or if the effective gain is zero. It still does not identify a biological carrier.

The finite-state kernel proves the exact Bellman solutions used by the adverse example. Bellman's name is not decorative: the recurrence separates the topology of possible routes from the probabilities and delays that determine expected arrival. The baseline first-step equation has the unique rational solution $h = 2$. The diverted two-state equations have the unique solution $h_0 = 2600$, $h_1 = 2650$. Lean separately proves that the baseline detour probability is zero while the diverted detour probability is positive, that each stated probability row is normalized, and that $2600 > 2$. Thus the support graph gains a route while the expected target time worsens under the changed kernel.

20.10 Adversarial mathematical boundaries

The formal review found five boundaries that must remain visible.

First, edge inclusion preserves routes but does not assign probabilities to them. Second, the favorable waiting-time theorem fixes the local process and adds an independent nonnegative clock; it cannot be applied when transfer changes mutation supply, state occupancy, competition, or the transition kernel. Third, exact carrier rescue is an algebraic identity in a specified response model, not evidence that a biological intervention is selective. Fourth, real-number equations do not carry physical units automatically; rate and response variables require a dimensional interpretation when fitted to data. Fifth, the adverse rational witness proves possibility, not prevalence.

Andrey Markov, Richard Bellman, Claude Shannon, Ronald Fisher, Sewall Wright, Alan Turing, Francis Crick, and Eric Kandel are useful here for different reasons. Markov supplies the transition baseline; Bellman supplies the first-step recurrence for expected future arrival; Shannon supplies the conditional-information question; Fisher and Wright prevent arrival from being confused with population fate; Turing legitimizes time-organized local dynamics as a source of larger pattern; Crick states the turnover problem; and Kandel supplies the signal-to-structure memory bridge. Their names add credibility only because each solves a named part of the argument. None is invoked as authority for the untested SET synthesis.

21 Experimental predictions and discriminators

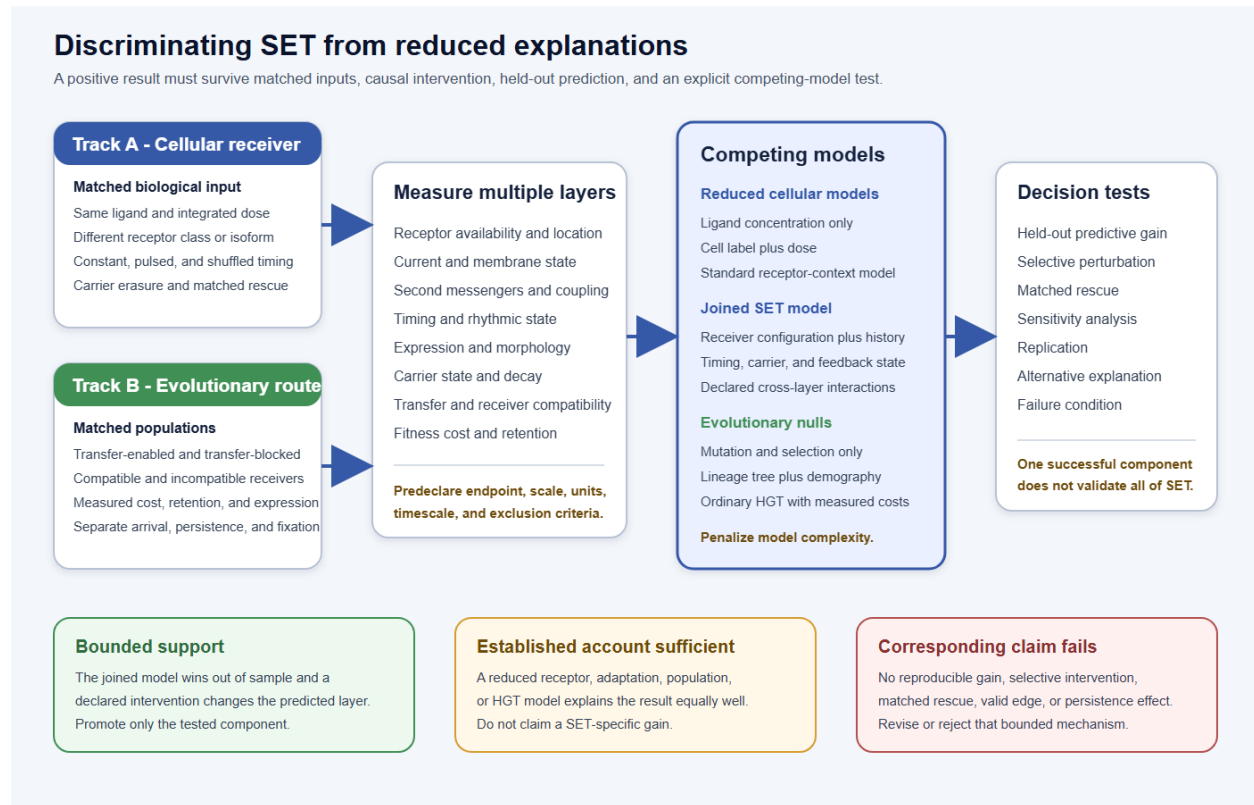


Figure 3: Two-track SET experiment architecture. Matched cellular and evolutionary inputs feed multi-layer measurements, competing models, held-out prediction, selective intervention, rescue, replication, and explicit failure conditions.

21.1 Prediction 1 - Receiver state outperforms ligand identity alone

Models incorporating receptor subtype, abundance, location, configuration, coupling, prior exposure, and current cell state should predict intracellular consequences better than ligand concentration alone.

Failure condition: the receiver variables add no reproducible held-out predictive value.

21.2 Prediction 2 - Same ligand, different receptor operation

In matched systems expressing ionotropic 5-HT₃ versus selected metabotropic receptors, serotonin should produce distinguishable fast-current and second-messenger trajectories predictable from receptor class.

Failure condition: receptor-class information does not improve response prediction beyond nonspecific cell labels.

21.3 Prediction 3 - Matched dose, different timing, different outcome

Temporally structured and shuffled inputs with matched dose and duration should produce different trafficking, second-messenger, expression, or morphological outcomes where SET claims timing is causal.

Failure condition: integrated dose explains the outcome and timing adds no effect.

21.4 Prediction 4 - Receiver editing or isoform changes routing

At matched expression, defined receptor isoforms or editing states should produce predicted differences in proximal coupling and downstream effector choice.

Failure condition: the configuration variable has no reproducible effect after expression and assay differences are controlled.

21.5 Prediction 5 - Signal termination changes integration windows

Manipulating autoreceptors, SERT, MAO, receptor internalization, or neighboring-cell uptake should change the interval over which sequential inputs interact.

Failure condition: measured clearance or availability changes do not alter the predicted integration window.

21.6 Prediction 6 - Oscillatory state gates an identified pathway

Perturbing electrical, calcium, metabolic, or transcriptional phase while controlling total input should alter a specified pathway or structural consequence.

Failure condition: the phase variable does not affect pathway accessibility after confounds are controlled.

21.7 Prediction 7 - A retained carrier explains matched-input history dependence

Cells exposed to controlled histories and then given a matched input should show a reproducible response difference predicted by a named receptor, metabolic, protein, morphological, chromatin, genetic, or transferred carrier. Blocking or erasing that carrier should reduce the history-separation statistic, and selective rescue should restore it where technically possible.

Failure condition: the effect is explained by present measured state, damage, fatigue, depletion, selection among pre-existing states, or nonspecific intervention; or no candidate carrier predicts erasure and rescue.

21.8 Prediction 8 - Transfer changes path statistics under defined compatibility

Matched transfer-enabled and transfer-suppressed microbial populations should differ in time to acquire compatible modules, module provenance, path length, or lineage dependence when donor contact, carrier route, receiver compatibility, expression, and retention are present.

Failure condition: transfer adds no predictive or measured advantage under the specified ecology, the necessary carrier or compatibility factor is absent, or apparent imports are independent reconstruction.

21.9 Prediction 9 - Hidden history defeats an impoverished state model

Populations with the same observed phenotype but different receptor, stress, or regulatory histories should produce different next-step distributions.

Failure condition: an observed-state memoryless model performs as well as the augmented model on held-out trajectories.

21.10 Prediction 10 - Receiver permissiveness predicts module persistence

Receivers with experimentally measured entry, replication, expression, and retention compatibility should preserve a defined module more reliably than matched incompatible receivers. Compensatory adaptation may increase persistence while changing portability.

Failure condition: the compatibility variables add no held-out predictive information after population size, ecology, and selection are controlled.

21.11 Prediction 11 - Transfer can accelerate, inhibit, or redirect adaptation

Across positive, negative, and absent selection regimes, a transfer route may lower first-arrival time, impose a cost, divert mutation supply, create intragenomic conflict, or enable reversal toward an ancestral state. The signed outcome should be predictable from the full rate-and-cost model rather than from transfer presence alone.

Failure condition: the expanded model does not outperform a baseline containing mutation, recombination, population structure, ecology, and selection.

21.12 Prediction 12 - Topological access does not determine expected adaptation time

When an added transfer route changes transition probabilities, receiver load, conflict, or ecological diversion, expanded reachability or a shorter nominal path need not lower expected hitting time. A measured transition model should predict the sign and magnitude of the change better than edge count alone.

Failure condition: after relevant transition rates and costs are measured, the richer model adds no predictive value over simple reachability or shortest-path statistics.

22 General failure conditions

SET must be narrowed or rejected in a claimed regime if:

1. receptor configuration and history add no predictive information beyond ligand, present cell state, and genotype;
2. the proposed receptor alphabet is redundant once cross-reactivity, saturation, and shared pathways are modeled;
3. timed or oscillatory state never changes pathway accessibility after dose and structure are controlled;
4. proposed persistent states decay before they influence action, selection, or heredity;
5. matched-input history effects cannot be tied to a retained carrier or survive carrier-specific erasure and rescue tests;
6. adaptation, habituation, or memory claims reduce to damage, depletion, saturation, sensitization, or selection among pre-existing states;
7. transfer routes do not alter reachability, waiting time, or module reuse;
8. imported modules cannot be distinguished from independent reconstruction;
9. existing models already contain the same state, timing, persistence, and topology operators, leaving SET without a distinct contribution;
10. the cellular-to-evolutionary bridge depends on causal steps that cannot be isolated;
11. a result applies only to a restricted microbial, neural, developmental, or pharmacological system but is presented as universal.

23 Intellectual and contribution history

SET emerged from the joining of two research paths.

Micah Blumberg's COT and SAN work supplied the micro-scale proposal: receptors and cellular structures act as state-dependent receivers; coincidence and time-structured differences alter downstream dynamics; rhythmic state can act as a control variable; and repeated signaling can become persistent structural bias.

Ezekiel Golan supplied or sharpened the macro-scale logistics and mathematical challenge: microbial horizontal gene transfer, plasmids, convergence probabilities, ergodicity, graph structure, and reusable modules moving across biological networks.

The named scientific landmarks perform defined jobs. Darwin supplies descent with modification and natural selection; Fisher, Haldane, and Wright supply population-genetic rate, selection, drift, structure, and landscape baselines; Kimura constrains selection-only narratives; Ohno supplies duplication; Jacob supplies reuse through evolutionary tinkering; King and Wilson supply regulatory evolution; Avery, MacLeod, and McCarty supply transformation; Lederberg and Tatum supply bacterial genetic exchange; and Zinder and

Lederberg supply transduction. Langley supplies receptive substances; Hodgkin and Huxley make electrical receiver state quantitative; Jacob and Monod connect conditions to gene regulation; Monod, Wyman, and Changeux formalize allosteric state; Waddington supplies developmental path dependence; Shannon supplies conditional information; and Turing connects temporal dynamics to spatial pattern. Koshland makes receptor adaptation experimentally tractable; Barkai and Leibler supply robustness; Bray supplies molecular computation; Hebb poses the coordinated-activity connection-change problem; Kandel joins signaling, genes, synapses, and structure; and Crick supplies the molecular-turnover constraint. Tsien, Lewis, and Lahav make timing-dependent transcription and fate measurable; Si, Kandel, Heinrich, and Lindquist supply bounded local-protein-synthesis and self-reinstating-protein mechanisms; and Netea and Stunnenberg supply a bounded chromatin-and-metabolic memory program. Markov defines a transition baseline; Bellman supplies the recurrence that converts transition probabilities into expected future arrival; McClintock supplies mobile elements; Margulis supplies evolutionary combination; Woese supplies an early communal regime; Doolittle establishes the limits of one universal tree; and Kunin, Dagan, and Martin supply explicit gene-sharing and modular-network reconstructions.

The receptor-specific experimental spine remains equally important. Twarog and Page locate serotonin in mammalian tissues including brain; Walther separates the central synthesis route; Katz and Miledi make release timing measurable; Geppert and Südhof identify a major fast-release calcium sensor; Heuser and Reese make vesicle reuse physical; Hare Bernheim anchors monoamine oxidation; Sutherland establishes second-messenger logic; Jensen supplies a contrasting intracellular-receptor architecture; and Nichols supplies an integrative map of psychedelic pharmacology. Burns, Berg, Julius, Wacker, Richardson-Jones, Trillat, Schmid, Vargas, Ippolito, and Beliveau supply receptor-specific tests. Each name is present because a defined result constrains SET; no name is used as prestige in place of evidence.

24 Scope and evidentiary boundary

The present theory spans COT, PWD, packetized evolution, the wormhole effect, hidden history, vibration-assisted transport, the receptor ecosystem, serotonin subtypes, the complete signal lifecycle, pharmacological perturbation, cellular learning, structural memory, evolutionary networks, executable models, and contribution history.

Its scientific and mathematical architecture includes:

- a modern evolutionary baseline separating mutation, recombination, duplication, regulation, transposition, transfer, selection, drift, and demographic history;
- defined scientific roles for Darwin, Fisher, Haldane, Wright, Kimura, Ohno, Jacob, King, Wilson, Avery, MacLeod, McCarty, Lederberg, Tatum, and Zinder;
- a receptor-family diversification bridge for the serotonin worked example;
- separate transformation, conjugation, transduction, transposition, and symbiosis mechanisms;
- eight required packet criteria;
- a successful-transfer rate containing contact, route, survival, receiver compatibility, expression, and retention;
- a typed multilayer evolutionary graph;
- separate equations for reachability, shortest path, first-arrival hazard, expected waiting time, and post-acquisition cost;
- favorable, adverse, and null transfer cases;
- explicit distinctions among arrival, persistence, fixation, and adaptive value;
- a declared evolutionary-network and module-reuse specification;

- a deterministic, inspectable synthetic model with a frozen seed;
- held-out receiver-state recovery against two reduced models;
- matched-dose temporal dynamics;
- explicit carrier erasure and rescue;
- analytic and Monte Carlo first-arrival checks;
- a bounded Wright-Fisher persistence simulation;
- graph-monotonicity checks and an adverse expected-hitting-time counterexample;
- a compatibility sensitivity CSV, a four-panel SVG figure, and nine automated tests.

Its formal layer adds:

- a pinned Lean 4.26.0 and Mathlib 4.26.0 proof package;
- fifteen machine-checked theorem statements;
- finite path, reachability, and path-bound preservation under edge extension;
- nonnegativity of the six-factor transfer rate;
- nonincrease and strict decrease theorems for independent-hazard waiting time;
- a composed theorem connecting the factored transfer rate to the wait bound;
- a positive response-domain theorem and three carrier-response identities;
- an explicit zero-to-positive detour-support theorem;
- unique exact solutions for the baseline and diverted Bellman equations;
- a machine-checked adverse hitting-time witness;
- a theorem-status ledger, claim signature, dependency lock, build receipt, and axiom/placeholder audit;
- an adversarial boundary separating graph support, stochastic dynamics, synthetic causality, and biological evidence.

The mathematical results remain conditional on their declared models. The biological claims remain answerable to measurement, intervention, replication, and comparison with reduced and rival explanations. Formal validity cannot substitute for biological parameter identification, and a biological observation cannot silently promote a bounded formal result into a universal theorem.

25 Conclusion

The same molecule can perform different biological work because signals are completed by receivers and extended through a lifecycle. Serotonin makes the principle visible: precursor supply constrains synthesis; VMAT2 prepares vesicular storage; calcium timing shapes release; diffusion and uptake shape reach; one receptor class directly conducts ions while others recruit distinct G-protein families; editing and conformation alter coupling; location changes access and circuit function; internalization changes future sensitivity; autoreceptors change future supply; and SERT, neighboring cells, MAO, receptor trafficking, and vesicle recycling determine how an episode ends.

Super Evolution Theory joins these operations into a receiver-centered architecture. The receptor ecosystem is the configurable input-and-control surface. COT names the proposed cellular reconstruction of actionable state from distributed, timed evidence. PWD names a receiver-relative departure capable of changing future processing. Pharmacological perturbations expose which stage or receiver variable matters. A response becomes a memory claim only when a controlled history is retained in an identified carrier, changes a matched later response, and survives erasure-and-rescue analysis. Packetized evolution names bounded biological content only when carrier, route, receiver, expression, retention, cost, consequence, selection, and drift are specified. The executable models make these distinctions calculable, and the formal kernel proves their

assumption-clean mathematical core while preserving the counterexample that limits it. The result is stronger precisely because it says both what follows and what does not: compatible independent transfer can lower first-arrival time, but graph expansion alone cannot establish faster expected evolution.

The names of Darwin, Fisher, Haldane, Wright, Kimura, Ohno, Jacob, McClintock, Margulis, Woese, Doolittle, Langley, Katz, Sutherland, Turing, Shannon, Waddington, and other scientific landmarks are not invoked as inherited authority. Each marks a specific problem SET must connect without erasing: receiving, release, intracellular transformation, pattern formation, information, path dependence, landscapes, mobile elements, and evolutionary combination. The scientific question is whether the joined architecture predicts interventions and histories better than reduced alternatives, whether its mathematical claims survive explicit assumptions, and whether experiment can distinguish SET from existing receptor, adaptation, evolutionary, and network models.

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